GREENS TECHNOLOGYS

ONE COURSE. AN END-TO-END DATA STACK.

AZURE DATA ENGINEERING

Build pipelines. Turn data into opportunity.



PLACEMENTS GUARANTEED

SQL | Python | Azure | ADF | Fabric | Synapse | Databricks

Azure Data Engineering Master Course

160 live sessions | 240 hours | 42 modules

Since 2008 - 5,00,000+ careers launched

From foundations to data pipelines

Learn the data stack step by step, from SQL and Python to cloud pipelines, a warehouse and business dashboards.

SQL + Python | Azure + ADF | Databricks + Fabric | Power BI

Build the foundations. Then connect the whole picture.

Extract data, transform it, load a warehouse and turn it into reports. Develop a project you can explain in interviews.

Who this course is for

Freshers | IT and non-IT graduates

Graduates from IT, engineering, arts, science, commerce and other degrees who want to start learning data engineering.

Non-IT employees | Moving into data

Working professionals changing careers who want to learn SQL, Python and cloud data skills from the foundations.

IT employees | Moving into data engineering

Developers, testers, support staff, database teams and cloud professionals who want to build data pipelines.

Data and analytics employees | Broadening their skills

SQL, BI, reporting and analytics professionals who want to add Azure, Databricks, Synapse and Fabric to their toolkit.

One course. A clear sequence.

Build confidence in the foundations, work through the cloud data stack and finish with an end-to-end capstone.

240 hours | 42 modules

160 live sessions | 4 capstone projects

01	Foundations (Modules 01-07)	37 sessions
02	Azure Core, Storage & Security (08-10)	12 sessions
03	Azure Data Factory (11-16)	18 sessions
04	Azure Databricks (17-24)	28 sessions
05	Azure Synapse Analytics (25-29)	13 sessions
06	Microsoft Fabric (30-35)	21 sessions
07	Streaming, Governance & DataOps (36-42)	17 sessions
08	Capstone projects	8 sessions
09	Certification prep & mock exams	4 sessions
10	Career readiness	2 sessions

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PHASE 1 OF 7 · MODULES 01–07 · 37 SESSIONS

Foundations

Pipelines, Python, SQL, PySpark, partitioning and ETL/ELT

01 MODULE 01 · Data Pipeline Fundamentals

3 SESSIONS | 33 TOPICS | Foundation ALIGNED

The vocabulary and mental models every data engineer uses daily, before any tool is opened. LAB: Whiteboard an end-to-end retail pipeline: sources, Bronze-Silver-Gold zones, star schema and the serving layer.

The Data Engineer Role (5)

- Data engineer vs analyst vs scientist
- Day-to-day responsibilities
- Producers, pipelines, consumers
- Data engineering lifecycle
- Modern data stack on Azure

Data Types & Formats (5)

- Structured data
- Semi-structured: JSON, XML
- Unstructured: logs, images, PDFs
- Row vs columnar storage
- CSV, JSON, Parquet, Avro, ORC, Delta

Data Modelling (6)

- Facts and dimensions
- Star schema
- Snowflake schema
- Grain of a fact table
- Surrogate vs natural keys
- Slowly Changing Dimensions intro

Data Systems (6)

- OLTP vs OLAP
- Data warehouse
- Data lake
- Data lakehouse
- Data mesh and data products
- Operational vs analytical stores

Pipeline Concepts (6)

- Ingest, store, process, serve
- Batch vs streaming
- Micro-batch processing
- Full vs incremental loads
- Idempotent pipelines
- Backfills and reprocessing

Architecture Patterns (5)

- Medallion: Bronze, Silver, Gold
- Lambda architecture
- Kappa architecture
- Data quality and SLAs
- Lineage and observability

02 MODULE 02 · GREENS CORE · Python & PySpark I: Python for Data Engineering

8 SESSIONS | 102 TOPICS | Foundation ALIGNED

Full Python from zero to production-style scripts, written the way data engineers actually use it. LAB: Build a Python ingestion script that pulls a paginated REST API, cleans records with Pandas, logs every step and writes partitioned JSON/CSV files.

Environment Setup (6)

- Installing Python
- Anaconda and virtual environments
- venv and pip
- requirements.txt
- Jupyter Notebook
- VS Code setup and extensions

Operators (7)

- Arithmetic operators
- Comparison operators
- Logical: and, or, not
- Assignment operators
- Membership: in, not in
- Identity: is, is not
- Operator precedence

Strings (7)

- Indexing and slicing
- String methods: split, join, strip, replace
- upper, lower, title, find
- f-strings and format()
- Escape characters
- Raw strings
- Regular expressions (re)

Basic Syntax & Data Types — 6

- Variables and naming rules
- Numbers, strings, booleans
- None type
- Type conversion and casting
- Input and print()
- Comments and docstrings

Control Flow — 8

- if, elif, else
- Nested conditions
- Ternary expressions
- for loops and range()
- while loops
- break, continue, pass
- Loop else clause
- match-case (structural pattern matching)

Data Structures — 6

- Lists and list methods
- Tuples and unpacking
- Sets and set operations
- Dictionaries and dict methods
- Nested structures
- List, dict and set comprehensions

03 MODULE 03 · GREENS CORE
SQL I: Core SQL for Data Engineering

6 SESSIONS | 65 TOPICS | DP-700 ALIGNED

Relational foundations and every SQL command category, practised on a real banking database. LAB: Design and normalise a banking schema (customers, accounts, transactions), load sample data, secure it with roles and answer 25 business queries.

Introduction to Relational Databases — 6

- Database concepts
- Tables, rows, columns
- Relationships: 1:1, 1:M, M:M
- Overview of popular SQL systems
- SQL Server, Azure SQL, PostgreSQL
- SSMS and Azure Data Studio

Data Types — 6

- Integer types
- Decimal and numeric
- Character: CHAR, VARCHAR, NVARCHAR
- Date and time types
- Boolean / BIT
- JSON support

DCL & TCL — 7

- GRANT, REVOKE, DENY
- Users, roles, schemas
- BEGIN TRANSACTION
- COMMIT and ROLLBACK
- SAVEPOINT
- ACID properties
- Isolation levels

Normalization & Keys — 7

- 1NF, 2NF, 3NF
- BCNF
- Denormalisation trade-offs
- Primary key
- Foreign key
- Unique, check, default, not null
- ER diagrams

DDL — 5

- CREATE DATABASE, SCHEMA, TABLE
- ALTER TABLE
- DROP vs TRUNCATE
- Computed columns
- Identity and sequences

SQL Fundamentals (DQL) — 8

- SELECT, FROM, WHERE
- ORDER BY
- DISTINCT
- TOP / OFFSET-FETCH
- LIKE, IN, BETWEEN
- IS NULL / COALESCE
- CASE expressions
- Aliases

DML — 5

- INSERT single and multi-row
- INSERT ... SELECT
- UPDATE with joins
- DELETE with conditions
- OUTPUT clause

04 MODULE 04 · GREENS CORE
SQL II: Advanced SQL & Warehouse Patterns

6 SESSIONS | 55 TOPICS | DP-700 ALIGNED

The advanced SQL interviewers ask for and warehouses run on: windows, CTEs, MERGE and SCD. LAB: Build a customer dimension with SCD Type 2 using MERGE, then write window-function reports for monthly running balances and top-3 products per region.

Window Functions (9)
- OVER() clause
- PARTITION BY and ORDER BY
- ROW_NUMBER, RANK, DENSE_RANK
- NTILE
- LAG and LEAD
- FIRST_VALUE, LAST_VALUE
- Running totals
- Moving averages
- Frame clauses: ROWS vs RANGE

Stored Procedures & Functions (6)
- Stored procedures with parameters
- Output parameters
- Scalar functions
- Inline table-valued functions
- Error handling: TRY...CATCH
- Triggers: AFTER and INSTEAD OF

Incremental Load Patterns (5)
- Watermark columns
- Change Data Capture (CDC)
- Change Tracking
- Late-arriving data
- Handling deletes

Indexes & Query Optimization (7)
- Clustered vs non-clustered index
- Covering indexes
- Columnstore indexes
- Execution plans
- Statistics
- SARGable predicates
- Best practices for query optimization

MERGE & Upserts (5)
- MERGE syntax
- WHEN MATCHED / NOT MATCHED
- NOT MATCHED BY SOURCE
- Upsert patterns
- Deduplication before merge

CTEs (6)
- Common Table Expressions
- Multiple CTEs
- CTEs for DML
- Recursive CTEs
- Hierarchies: org charts
- CTE vs subquery vs temp table

Slowly Changing Dimensions (6)
- SCD Type 0 and Type 1
- SCD Type 2: history rows
- Effective dates and is_current flag
- SCD Type 3
- SCD Type 4 and 6 overview
- Implementing SCD2 with MERGE

Interview Problem Sets (5)
- Nth highest salary
- Consecutive days / gaps and islands
- Top-N per group
- Duplicate detection
- Year-over-year growth

Advanced Query Techniques (6)
- Views and indexed views
- Temp tables: local and global
- Table variables
- PIVOT and UNPIVOT
- APPLY: CROSS and OUTER
- Dynamic SQL basics

05 MODULE 05 · GREENS CORE
Python & PySpark II: PySpark Foundations

7 SESSIONS | 68 TOPICS | DP-700 ALIGNED

Distributed processing with the DataFrame API, from SparkSession to production-ready transformations. LAB: Ingest nested banking JSON with an explicit schema, flatten it, mask PAN numbers, join with customers and write partitioned Parquet.

Introduction to PySpark (5)
- Why distributed computing
- Hadoop, HDFS and MapReduce in brief
- The role of PySpark in big data
- Spark vs Pandas
- Spark ecosystem: SQL, Streaming, MLlib

DataFrames & Schemas (6)
- Creating DataFrames
- Schema inference
- StructType and StructField
- Explicit schemas for production
- printSchema and describe
- Column objects and col()

Window Functions in PySpark (5)
- Window.partitionBy().orderBy()
- row_number, rank, dense_rank
- lag and lead
- Running totals
- rowsBetween and rangeBetween

Complex & Semi-Structured Data (5)
- Arrays and explode
- Structs and nested fields
- Maps
- from_json and to_json
- Flattening nested JSON

Reading & Writing Data (7)
- CSV options: header, delimiter, mode
- JSON and multiline JSON
- Parquet and Avro
- Delta format
- JDBC sources
- Write modes: append, overwrite, ignore
- partitionBy on write

Spark Architecture (6)
- Driver and executors
- Cluster manager
- Jobs, stages, tasks
- DAG and lazy evaluation
- Transformations vs actions
- Narrow vs wide transformations

Functions & UDFs (6)
- String functions
- Date and timestamp functions
- Null handling: fillna, dropna, coalesce
- Python UDFs
- Pandas (vectorised) UDFs
- Why built-ins beat UDFs

Setting Up PySpark (5)
- Local install and PySpark in Jupyter
- Databricks Free Edition
- SparkSession and its importance
- SparkContext
- Spark configuration basics

Core Transformations (7)
- select and selectExpr
- filter / where
- withColumn and withColumnRenamed
- drop and dropDuplicates
- cast and data types
- when / otherwise
- lit and expr

Spark SQL (5)
- createOrReplaceTempView
- Global temp views
- spark.sql() queries
- DataFrame API vs Spark SQL
- Catalog and tables

RDDs (5)
- Resilient Distributed Datasets
- Creating RDDs
- map, flatMap, filter
- reduceByKey
- When to still use RDDs

Joins & Aggregations (6)
- Inner, left, right, full joins
- Left semi and left anti joins
- Cross join
- groupBy and agg
- pivot
- union and unionByName

06 MODULE 06 · PySpark Performance & Spark Internals

4 SESSIONS | 36 TOPICS | DP-700 ALIGNED

Why jobs are slow and how to fix them: the tuning knowledge that separates juniors from seniors. LAB: Take a deliberately slow join job, diagnose it in the Spark UI and cut its runtime using broadcast, salting, AQE and right-sized partitions.

Reading the Spark UI (5)

- Jobs, stages and tasks tabs
- SQL / DataFrame tab
- Executors tab
- Spotting skew and spill
- Event timeline

Query Plans (6)

- explain() modes
- Logical vs physical plan
- Catalyst optimizer
- Tungsten engine
- Predicate pushdown
- Column pruning

Shuffles & Joins (6)

- What causes a shuffle
- spark.sql.shuffle.partitions
- Broadcast hash join
- Sort-merge join
- Join hints
- Bucketing

07 MODULE 07 · Partitioning, File Formats & ETL/ELT Patterns

3 SESSIONS | 33 TOPICS | DP-700 ALIGNED

Designing data layouts and load patterns that stay fast and cheap as data grows. LAB: Partition one year of transactions three different ways, measure query time and file counts, and recommend a layout.

Partitioning Strategies (6)

- Why partition
- Partition by date: year/month/day
- Hive-style partition folders
- Choosing partition columns
- Over-partitioning pitfalls
- Partition pruning

File Formats Deep Dive (5)

- Parquet internals: row groups, column chunks
- Avro for streaming
- Delta Lake format
- Apache Iceberg overview
- Compression: Snappy, ZSTD, GZIP

Load Patterns (6)

- Full load
- Incremental append
- Upsert / merge
- CDC-based loads
- Snapshot vs delta
- Late-arriving data handling

ETL vs ELT (5)

- ETL pattern
- ELT pattern
- When to choose which
- Push-down transformations
- Reverse ETL

Beyond Partitioning (5)

- Z-ordering
- Liquid clustering
- V-Order in Fabric
- Bucketing
- Data skipping and statistics

Pipeline Design Principles (6)

- Idempotency
- Restartability and checkpoints
- Metadata-driven design
- Data contracts
- Data quality gates
- Error and quarantine tables

2

PHASE 2 OF 7 · MODULES 08–10 · 12 SESSIONS

Azure Core, Storage & Security

Cloud fundamentals, ADLS Gen2, Entra ID, RBAC, Key Vault

08

MODULE 08 · GREENS CORE

Azure Cloud Fundamentals

5 SESSIONS | 50 TOPICS | Foundation ALIGNED

The Azure building blocks every data platform sits on, set up hands-on in your own subscription. LAB: Create a governed resource group with tags, a storage account via CLI, an Azure SQL database and a Logic App that emails on new files.

Overview of Azure Architecture (8)

- Cloud models: IaaS, PaaS, SaaS
- Azure regions and availability zones
- Region pairs
- Tenants, management groups, subscriptions
- Resource groups and tagging
- Azure service categories
- Designing scalable and secure cloud solutions
- Azure Well-Architected Framework

Entra ID (formerly Azure Active Directory) (5)

- Overview of identity and access management
- Users, groups, app registrations
- Role-Based Access Control (RBAC)
- Authentication flows: OAuth 2.0, OIDC
- Multi-factor authentication

Azure SQL (6)

- Azure SQL Database
- SQL Managed Instance
- SQL on Azure VMs
- Purchasing models: DTU vs vCore
- Serverless tier
- Performance, scalability and security features

Virtual Network (VNet) (6)

- Fundamentals of virtual networking
- Subnets and IP addressing
- Network security groups
- Private endpoints
- Service endpoints
- VNet peering basics

Logic Apps (4)

- Building workflows without code
- Connectors and triggers
- Integrating Azure and external services
- Email and Teams alerts from pipelines

Azure Tooling (5)

- Azure Portal navigation
- Azure CLI
- Azure PowerShell
- Cloud Shell
- Azure Resource Manager (ARM)

Azure Key Vault Services (4)

- Secure storage for keys, secrets, certificates
- Access policies vs RBAC
- Integrating Key Vault with other services
- Secret rotation

Azure Functions (5)

- Serverless and event-driven computing
- Triggers and bindings
- Creating and deploying Functions
- Scaling and hosting plans
- Functions in data pipelines

Blob Storage (7)

- Storage account types
- Containers and blobs
- Block, append, page blobs
- Access tiers: hot, cool, cold, archive
- Redundancy: LRS, ZRS, GRS, GZRS
- Lifecycle management policies
- Soft delete and versioning

09 MODULE 09 · GREENS CORE
Azure Data Lake Storage (ADLS Gen2)

4 SESSIONS | 31 TOPICS | DP-700 ALIGNED

Building a secure, well-organised enterprise data lake that every downstream engine can share. LAB: Build a Medallion lake with ACLs per team, a SAS for a vendor, lifecycle rules to cool and archive, and a Python uploader.

Introduction to Azure Data Lake Storage (4)

- What a data lake is
- ADLS Gen1 vs Gen2 (Gen1 retired Feb 2024)
- Use cases in modern architectures
- ADLS Gen2 vs Blob Storage

Provisioning & Configuring ADLS (5)

- Creating an ADLS account
- Hierarchical namespace
- Containers, directories, files
- ABFS driver and abfss:// paths
- Azure Storage Explorer

Lake Design (4)

- Raw / Bronze, Silver, Gold zones
- Folder and naming conventions
- Landing and archive zones
- Separate lakes per environment

3

PHASE 3 OF 7 · MODULES 11–16 · 18 SESSIONS

Azure Data Factory

Orchestration, ingestion, transformation and operations

10 MODULE 10 · Identity, Secrets & Network Security: Entra ID, RBAC, Key Vault

3 SESSIONS | 25 TOPICS | DP-700 ALIGNED

Securing data platforms end to end: who can access what, with which identity, over which network. LAB: Replace every connection string in a pipeline with managed identity and Key Vault, then lock storage behind a private endpoint.

Entra ID in Depth (5)

- Service principals
- System vs user-assigned managed identities
- App registrations and secrets
- Conditional access overview
- Privileged Identity Management (PIM)

Key Vault for Data Engineers (5)

- Secrets for connection strings
- Key Vault in ADF linked services
- Databricks secret scopes backed by Key Vault
- Fabric connections and Key Vault
- Auditing secret access

Azure Policy & Compliance (4)

- Azure Policy basics
- Enforcing tags and regions
- Microsoft Defender for Cloud overview
- Compliance frameworks: GDPR, HIPAA, DPDP Act

RBAC Design (6)

- Built-in vs custom roles
- Scope: management group to resource
- Owner, Contributor, Reader
- Data-plane vs control-plane roles
- Least privilege
- Access reviews

Network Security (5)

- Private endpoints for storage and SQL
- Managed virtual networks in ADF
- Databricks VNet injection
- Fabric managed private endpoints
- Public access disablement

11 MODULE 11 · GREENS CORE ADF 1: Introduction to Azure Data Factory

2 SESSIONS | 14 TOPICS | DP-700 ALIGNED

Where ADF fits in the Azure ecosystem and what it is (and is not) the right tool for. LAB: Provision ADF, connect it to Git, and run a first copy pipeline from Blob to ADLS Gen2.

Overview of ADF (5)

- Role of ADF within the Azure ecosystem
- Key components: pipelines, activities
- Datasets and linked services
- Triggers
- ADF Studio tour

Use Cases & Scenarios (5)

- Batch data processing
- Data ingestion and migration
- ETL and ELT processes
- Orchestration with other Azure services
- Hybrid on-prem to cloud

ADF Family (4)

- ADF vs Synapse Pipelines
- ADF vs Fabric Data Factory
- ADF vs Databricks Jobs
- Pricing model: activity runs, DIUs

12 MODULE 12 · GREENS CORE
ADF 2: Core Concepts & Architecture

4 SESSIONS | 31 TOPICS | DP-700 ALIGNED

Pipelines, activities and expressions: the building blocks of every reusable ADF solution. LAB: Build one parameterised pipeline that copies any table passed to it, driven by a tumbling-window trigger.

Data Integration & Orchestration (3)

- How ADF moves and transforms data
- Code-free data flows vs code-based transforms
- Control flow vs data flow

Control Flow Activities (8)

- Lookup
- Get Metadata
- ForEach (sequential and batch)
- If Condition and Switch
- Until and Wait
- Set Variable and Append Variable
- Execute Pipeline
- Fail activity

Parameters & Expressions (6)

- Pipeline parameters
- Global parameters
- Variables
- Dynamic content
- Expression functions: concat, formatDateTime, utcnow
- System variables

ADF Pipeline Structure (4)

- Components of a pipeline
- Activity dependencies: success, failure, completion, skipped
- Copy, Data Flow, Stored Procedure activities
- Notebook, Script, Web activities

Triggers (5)

- Schedule trigger
- Tumbling window trigger
- Storage event trigger
- Custom event trigger
- Trigger parameters

Linked Services & Datasets (5)

- Defining connections to sources and sinks
- Configuring datasets
- Parameterised linked services
- Parameterised datasets
- Key Vault-backed secrets

13 MODULE 13 · GREENS CORE
ADF 3: Data Ingestion & Movement

4 SESSIONS | 26 TOPICS | DP-700 ALIGNED

Moving data reliably from any source, on-premises or cloud, at any volume. LAB: Build a metadata-driven framework that incrementally ingests 10 SQL tables and a paginated REST API into ADLS Bronze with audit logging.

Copy Activity — 6

- Configuring and optimising Copy Activity
- Structured, semi-structured, unstructured data
- Column mapping and type conversion
- Fault tolerance: skip incompatible rows
- Staged copy
- Copy Data tool

Connecting to Diverse Data Sources — 6

- On-premises SQL Server
- Cloud storage and databases
- SaaS applications
- REST APIs with pagination rules
- SFTP and file shares
- Authentication: managed identity, service principal

Metadata-Driven Framework — 5

- Config table design
- Lookup + ForEach pattern
- Dynamic source and sink
- Auditing and logging table
- Scaling to hundreds of tables

Incremental Ingestion — 4

- High-watermark pattern
- Control tables
- Change Data Capture resource
- Copy new files by last modified date

Integration Runtime (IR) — 4

- Azure IR
- Self-hosted IR
- Azure-SSIS IR (lift-and-shift of SSIS)
- Managed VNet IR
- When and how to use each type

14 MODULE 14 · GREENS CORE
ADF 4: Data Transformation Techniques

3 SESSIONS | 23 TOPICS | DP-700 ALIGNED

Transforming data visually with Mapping Data Flows, or handing off to Spark when logic gets heavy. LAB: Build an SCD Type 2 dimension in a Mapping Data Flow, then rebuild it as a Databricks notebook called from ADF.

Mapping Data Flows — 4

- Designing transformation logic visually
- Source and sink settings
- Debug mode and data preview
- Data flow parameters

Transformation Components — 9

- Select, Filter, Derived Column
- Join, Lookup, Exists
- Aggregate, Pivot, Unpivot
- Conditional Split
- Union
- Window
- Alter Row for upserts
- Flatten for JSON
- Surrogate Key

Performance in Data Flows — 4

- Cluster sizing
- Partitioning options
- Broadcast settings
- Performance considerations and tuning

Custom Transformations — 6

- Azure Databricks notebook activity
- Passing parameters and returning values
- Synapse notebooks and SQL
- Azure Functions activity
- Custom activity
- Stored procedure activity

15 MODULE 15 · GREENS CORE
ADF 5: Monitoring, Logging & Debugging

2 SESSIONS | 14 TOPICS | DP-700 ALIGNED

Operating pipelines in production: finding failures fast and alerting before users notice. LAB: Send ADF diagnostics to Log Analytics, write KQL for failed runs, and alert to email and Teams on failure.

ADF Monitoring Tools (5)
- Monitoring dashboard: pipeline and activity runs
- Gantt view
- Rerun from failed activity
- Alerts and notifications for failures
- Bottleneck alerts

Error Handling (4)
- Retry and timeout settings
- Try-catch-finally patterns
- Fail activity with custom messages
- Email and Teams alerts via Logic Apps

Logging & Diagnostics (5)
- Diagnostic settings for detailed logs
- Azure Monitor integration
- Log Analytics workspace
- Centralised logging and alerting
- KQL queries on ADF logs

16 MODULE 16 · GREENS CORE
ADF 6: Performance Tuning, Optimization & CI/CD

3 SESSIONS | 17 TOPICS | DP-700 ALIGNED

Making pipelines faster and cheaper, then promoting them safely from Dev to Prod. LAB: Tune a slow copy by 3x, then deploy the factory Dev → UAT → Prod with an Azure DevOps pipeline.

Optimizing Data Movement (5)
- Tuning parallelism and DIUs
- Batch sizes
- Performance settings
- Scaling Integration Runtimes
- Partitioned copy from databases

Optimizing Data Flows (4)
- Partitioning and caching strategies
- Monitoring resource usage
- Reducing latency, maximising throughput
- Cost control

CI/CD for ADF (5)
- ARM template export
- Automated publish with npm package
- Azure DevOps release pipelines
- Environment parameter files
- Pre and post-deployment scripts

Source Control (3)
- Git integration: Azure Repos, GitHub
- Collaboration and publish branches
- Feature branches

4

PHASE 4 OF 7 · MODULES 17–24 · 28 SESSIONS

Azure Databricks

Lakehouse, Delta Lake, Unity Catalog, Lakeflow

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MODULE 17 · GREENS CORE

Databricks 1: Databricks Platform

3 SESSIONS | 20 TOPICS | DEA ALIGNED

Getting fluent in the Databricks workspace, compute options and day-one tooling. LAB: Create compute with a cluster policy, run a parameterised notebook with widgets and schedule it as a job.

Overview (4)

- Databricks environment and core capabilities
- Azure Databricks architecture: control vs compute plane
- Collaborative workspace and dev tools
- Data Intelligence Platform overview

Cluster Management (8)

- All-purpose vs job compute
- Serverless compute
- SQL warehouses
- Access modes: standard and dedicated
- Databricks Runtime versions
- Autoscaling and cluster sizing
- Cluster policies
- Cost optimisation best practices

Jobs & Workflows (3)

- Building and scheduling jobs
- Orchestrating notebooks, jobs and pipelines
- Task values

Workspace Essentials (5)

- Notebooks and magic commands: %sql, %python, %md, %run
- dbutils: fs, secrets, widgets
- Widgets for parameters
- Workspace files and Git folders
- Databricks CLI

18

MODULE 18 · GREENS CORE

Databricks 2: Apache Spark Concepts on Databricks

3 SESSIONS | 12 TOPICS | DEA ALIGNED

Applying Spark at platform scale with Databricks-specific engines and features. LAB: Port a Pandas job to Spark on Databricks and compare with Photon on and off.

Spark Architecture (3)

- Execution model: driver, executors, tasks, partitions
- Distributed computing fundamentals
- Photon engine

Spark SQL & DataFrames (5)

- Spark SQL for querying data
- Transformations and actions
- DataFrames (Datasets are Scala-only)
- Higher-order functions
- SQL scripting and parameters

Spark Data Processing (4)

- Batch processing of large datasets
- Best practices for optimising Spark applications
- Pandas API on Spark
- Spark Connect basics

19 MODULE 19 · GREENS CORE
Databricks 3: Delta Lake & the Data Lakehouse

4 SESSIONS | 25 TOPICS | DEA ALIGNED

The storage layer behind the lakehouse: reliable, versioned and fast tables on ADLS. LAB: Build a customer SCD2 table with MERGE, break it, time-travel to fix it, then OPTIMIZE and VACUUM.

Introduction to Delta Lake (5)

- Fundamentals and role in modern architectures
- ACID transactions
- Schema enforcement
- _delta_log transaction log
- Checkpoint files

Implementing a Lakehouse (4)

- Lakes + warehouses combined
- Robust ETL pipelines with Delta
- Managed vs external tables
- CTAS and DEEP / SHALLOW CLONE

DML on Delta (5)

- INSERT, UPDATE, DELETE
- MERGE INTO for upserts
- SCD Type 1 with MERGE
- SCD Type 2 with MERGE
- Deletion vectors

20 MODULE 20 · GREENS CORE
Databricks 4: Data Ingestion & Integration

4 SESSIONS | 24 TOPICS | DEA ALIGNED

Getting batch and streaming data into the lakehouse incrementally and exactly once. LAB: Stream files with Auto Loader and events from Event Hubs into Bronze Delta with schema evolution and checkpointing.

Ingestion Techniques (5)

- Batch ingestion best practices
- Structured, semi-structured, unstructured data
- COPY INTO
- Unity Catalog volumes for files
- Lakeflow Connect managed connectors

Auto Loader (7)

- cloudFiles source
- File notification vs directory listing
- Schema location and inference
- Schema evolution modes
- Rescued data column
- Checkpoints
- Trigger: availableNow vs processingTime

Streaming Data 7

- Spark Structured Streaming
- Event time and processing time
- Watermarks
- Tumbling, sliding, session windows
- Output modes
- Exactly-once guarantees
- foreachBatch for MERGE

Integrating with External Systems 5

- ADLS Gen2 via external locations
- Azure SQL and JDBC
- Azure Event Hubs (Kafka endpoint)
- Lakehouse Federation
- Other clouds: S3 and GCS overview

21 MODULE 21 · GREENS CORE
Databricks 5: ETL Pipelines & Data Transformation

4 SESSIONS | 24 TOPICS | DEA ALIGNED

Production pipelines with Lakeflow: declarative, tested and orchestrated. LAB: Build a Bronze-Silver-Gold Lakeflow pipeline with expectations and SCD2, orchestrated by a multi-task job.

Designing ETL/ELT Workflows 3

- Efficient extract, transform, load strategies
- Modular pipeline design for reuse
- Medallion implementation on Databricks

Lakeflow Spark Declarative Pipelines (formerly DLT) 7

- Streaming tables
- Materialized views
- Expectations: warn, drop, fail
- AUTO CDC / APPLY CHANGES
- SCD Type 1 and 2 declaratively
- Pipeline modes: triggered vs continuous
- Event log and lineage

Lakeflow Jobs (Workflows) 6

- Multi-task jobs and dependencies
- If/else and For each tasks
- Job parameters
- Retries, timeouts, notifications
- File arrival and table triggers
- Calling jobs from ADF

Data Cleaning & Enrichment 4

- Cleaning and enriching raw data
- Spark functions and UDFs
- Deduplication patterns
- Standardising dates and types

Data Quality 4

- Quality rules and thresholds
- Quarantine tables
- Row count and null checks
- Lakehouse monitoring overview

22 MODULE 22 · GREENS CORE
Databricks 6: Performance Tuning & Optimization

3 SESSIONS | 18 TOPICS | DEA ALIGNED

Keeping Databricks fast and cost-efficient as data and users grow. LAB: Use Query Profile and system billing tables to cut a Gold pipeline's cost and runtime.

Optimizing Spark Jobs — 4

- Tuning Spark configurations
- Managing memory and resource allocation
- Partitioning strategies
- Photon and serverless trade-offs

Query Optimization — 5

- Writing efficient Spark SQL
- Caching and disk cache
- Broadcast joins
- Query plan analysis and Query Profile
- AQE in practice

Cost Control — 5

- Choosing compute types
- Auto-termination
- Spot instances
- System tables for billing
- Budgets and tagging

Storage Optimization — 4

- File sizing and auto compaction
- Liquid clustering vs partitioning
- Data skipping
- Predictive optimisation

23 MODULE 23 · GREENS CORE
Databricks 7: Data Security, Governance & Compliance

4 SESSIONS | 26 TOPICS | DEA ALIGNED

Governing every table, file, model and user with Unity Catalog, and proving compliance. LAB: Set up catalogs per environment, mask PAN columns, apply region row filters and trace lineage Bronze → Gold.

Secret Scope — 4

- Managing credentials with secret scopes
- Azure Key Vault-backed scopes
- dbutils.secrets usage
- Redaction in notebooks

Fine-Grained Access — 5

- Row filters
- Column masks
- ABAC and governed tags
- Dynamic views
- Groups and service principals

Legacy & Migration — 3

- DBFS mounts (deprecated) vs external locations
- Hive metastore to Unity Catalog migration
- UCX overview

Regulatory Compliance (GDPR, HIPAA, etc.) — 4

- Compliant processing and storage
- Right to be forgotten in Delta
- PII discovery and tagging
- Best practices for auditability and privacy

Unity Catalog — 4

- Centralised metadata and governance
- Metastore, catalog, schema, table
- Storage credentials and external locations
- Volumes
- GRANT and REVOKE
- Data lineage
- Access control policies

Sharing & Auditing — 4

- Delta Sharing
- System tables: audit, lineage, billing
- Unity Catalog metrics overview

5

PHASE 5 OF 7 · MODULES 25–29 · 13 SESSIONS

Azure Synapse Analytics

SQL pools, Spark pools and migration to Fabric

24 MODULE 24 · Databricks 8: DataOps, SQL & AI on Databricks

3 SESSIONS | 14 TOPICS | DEA ALIGNED

Shipping Databricks code like software and serving the results to analysts and AI. LAB: Package the Lakeflow pipeline as a bundle, deploy via CI/CD and publish a Genie space over Gold.

CI/CD on Databricks (5)

- Git folders and branching
- Declarative Automation Bundles (formerly Asset Bundles)
- databricks.yml targets: dev, prod
- Deploying with Azure DevOps or GitHub Actions
- Unit tests for notebooks

Databricks SQL (5)

- SQL warehouses: serverless vs pro
- SQL editor and queries
- AI/BI dashboards
- Alerts
- Power BI connection

AI on Databricks (4)

- AI/BI Genie spaces
- AI functions in SQL
- Databricks Assistant for code
- Vector search overview

25 MODULE 25 · GREENS CORE · Synapse 1: Azure Synapse Analytics Essentials

2 SESSIONS | 11 TOPICS | Legacy estate ALIGNED

Understanding Synapse, which still powers many enterprise warehouses, and where it is heading. LAB: Create a Synapse workspace on your ADLS and tour every Studio hub.

Overview & Architecture (4)

- Evolution from traditional DWH to Synapse
- Key components: SQL pools, Spark pools
- Pipelines and Synapse Studio
- MPP architecture: control and compute nodes

Integration (4)

- Connecting Synapse with ADLS and other services
- Workspace setup and management
- Linked services
- Managed identity access

Platform Direction (3)

- Synapse vs Databricks vs Fabric
- Microsoft's guidance to choose Fabric for new work
- Retired features to avoid: Data Explorer pools, Synapse Link for Cosmos DB in new projects

26 MODULE 26 · GREENS CORE
Synapse 2: Data Integration & Pipeline Orchestration

2 SESSIONS | 11 TOPICS | Legacy estate ALIGNED

Synapse Pipelines for batch orchestration inside the workspace. LAB: Build a copy → Spark transform → SQL load pipeline entirely in Synapse Studio.

Data Ingestion (4)
- Building and configuring Synapse Pipelines
- Copy activities
- Parameterisation and dynamic content
- Batch ingestion (streaming belongs to Event Hubs / RTI)

Workflow Automation (4)
- Integrating with external sources
- Orchestrating multi-step data flows
- Notebook and SQL script activities
- Triggers

Synapse Pipelines vs ADF (3)
- Feature differences
- When each is used
- Migration path to Fabric Data Factory

27 MODULE 27 · GREENS CORE
Synapse 3: Data Warehousing with SQL Pools

4 SESSIONS | 21 TOPICS | Legacy estate ALIGNED

Designing and loading MPP warehouses and querying the lake directly with T-SQL. LAB: Load a star schema into a dedicated pool with COPY INTO and compare it with a serverless logical warehouse.

SQL Pools Management (4)
- Provisioning dedicated and serverless SQL pools
- DWUs and scaling
- Pause and resume
- Cost management and performance best practices

Loading Data (4)
- COPY INTO
- PolyBase external tables
- CTAS patterns
- Staging tables

Advanced T-SQL & Data Modelling (5)
- Complex queries and stored procedures
- Materialized views
- Star and snowflake schemas
- Indexing strategies
- Result set caching

Serverless SQL (4)
- OPENROWSET on CSV, Parquet, Delta
- External tables and data sources
- Logical data warehouse
- Pay-per-query cost

Dedicated Pool Design (4)
- Distributions: HASH, ROUND_ROBIN, REPLICATE
- Choosing distribution keys
- Clustered columnstore indexes
- Partitioning in dedicated pools

28 MODULE 28 · GREENS CORE
Synapse 4: Big Data Processing with Spark Pools

2 SESSIONS | 8 TOPICS | Legacy estate ALIGNED

Running PySpark inside Synapse for lake transformations. LAB: Transform Bronze data to Silver Delta in a Spark pool and query it from serverless SQL.

Spark Pools & PySpark (4)
- Launching and configuring Spark pools
- Autoscale and auto-pause
- Synapse notebooks
- Library management

Data Transformations (4)
- DataFrame operations
- UDFs and transformations
- Writing Delta tables
- Lake database and shared metadata

29 MODULE 29 · GREENS CORE
Synapse 5: Performance Optimization, Security & Governance

3 SESSIONS | 21 TOPICS | Legacy estate ALIGNED

Tuning, securing and monitoring Synapse, then planning its move to Fabric. LAB: Apply RLS, column masking and workload groups, then draft a Synapse-to-Fabric migration plan.

Performance Tuning (4)
- Optimising SQL with indexing, partitioning, caching
- Statistics
- Data movement and skew
- Spark job tuning

Security & Compliance (6)
- RBAC and Synapse roles
- Data encryption (TDE)
- Network security and managed VNet
- Row-level security
- Column-level security
- Dynamic data masking

Migration to Fabric (4)
- Migration assistant overview
- Dedicated pool → Fabric Warehouse
- Spark pool → Fabric Spark
- Pipelines → Fabric Data Factory

Workload Management (4)
- Workload groups
- Classifiers
- Importance
- Monitoring DMVs and killing sessions

Monitoring (3)
- Azure Monitor for system health
- Alerts for proactive management
- Synapse monitor hub

6

PHASE 6 OF 7 · MODULES 30–35 · 21 SESSIONS

Microsoft Fabric

OneLake, Lakehouse, Warehouse, Data Factory, Real-Time Intelligence

30 MODULE 30 · Fabric 1: Microsoft Fabric Foundations & OneLake

3 SESSIONS | 19 TOPICS | DP-700 ALIGNED

Microsoft's unified SaaS analytics platform and the focus of the current DP-700 exam. LAB: Create a Fabric workspace in a domain, configure Spark settings and add shortcuts to your ADLS Gold zone.

Fabric Overview (5)

- What Fabric is and why it exists
- Workloads: Data Engineering, Data Factory, Data Warehouse, Real-Time Intelligence, Power BI
- Capacities and SKUs
- Fabric trial
- Copilot in Fabric

OneLake (5)

- One lake per tenant
- Delta Parquet as the common format
- OneLake shortcuts: ADLS, S3, Dataverse
- OneLake file explorer
- OneLake catalog

Workspaces & Domains (6)

- Creating and organising workspaces
- Domain workspace settings
- Workspace roles: Admin, Member, Contributor, Viewer
- Spark workspace settings
- OneLake workspace settings
- Apache Airflow workspace settings

Choosing the Right Item (3)

- Lakehouse vs Warehouse vs Eventhouse
- Dataflow Gen2 vs pipeline vs notebook
- Fabric SQL database overview

31 MODULE 31 · Fabric 2: Lakehouse & Data Engineering

4 SESSIONS | 21 TOPICS | DP-700 ALIGNED

Spark-based engineering in Fabric: notebooks, Delta tables and the SQL analytics endpoint. LAB: Build a Medallion lakehouse in Fabric with notebooks, incremental loads and optimised Gold tables.

Fabric Lakehouse
5

- Tables vs Files sections
- SQL analytics endpoint
- Default semantic model
- Lakehouse schemas
- Loading data via upload, pipeline, notebook

Transforming Data
6

- PySpark and Spark SQL in Fabric
- Full and incremental loads
- Preparing data for a dimensional model
- Denormalising data
- Group and aggregate
- Handling duplicate, missing and late- arriving data

Optimising Lakehouse Tables
5

- V-Order
- Optimize write
- OPTIMIZE and VACUUM
- Table maintenance jobs
- Materialized lake views

Notebooks & Spark
5

- Fabric notebooks
- Spark pools and environments
- Starter pools and high concurrency
- notebookutils (mssparkutils)
- Parameters and notebook orchestration

32 MODULE 32 · Fabric 3: Data Warehouse & T- SQL

3 SESSIONS | 17 TOPICS | DP-700 ALIGNED

A fully managed T-SQL warehouse on OneLake, with no indexes or distributions to manage. LAB: Load a star schema into a Fabric Warehouse with stored procedures and SCD2, then tune it with query insights.

Fabric Warehouse
5

- Warehouse vs SQL analytics endpoint
- Creating tables and loading data
- COPY INTO and CTAS
- Cross-database queries
- Zero-copy table clones

T-SQL Transformations
4

- Stored procedures for ELT
- MERGE and SCD in the warehouse
- Views and functions
- Time travel queries

Optimising a Data Warehouse
5

- Statistics
- Query insights
- Result set caching
- Data clustering
- Workload management

Database Projects
3

- SQL database projects
- Schema as code
- Deploying with VS Code

33 MODULE 33 · Fabric 4: Data Factory in Fabric

4 SESSIONS | 21 TOPICS | DP-700 ALIGNED

Ingestion and orchestration in Fabric: pipelines, Dataflow Gen2, copy jobs, Airflow and mirroring. LAB: Mirror an Azure SQL database, ingest a REST API with a pipeline and clean files with Dataflow Gen2.

Fabric Pipelines (5)

- Pipeline activities
- Parameters and dynamic expressions
- Schedules and event-based triggers
- Orchestration patterns with notebooks and pipelines
- Connections and gateways

Dataflow Gen2 (5)

- Power Query Online
- Transformations without code
- Data destinations
- Fast copy
- Incremental refresh

Apache Airflow Job (3)

- Airflow concepts: DAGs, tasks, operators
- Running Airflow in Fabric
- When to choose Airflow

Mirroring (5)

- Mirroring concepts
- Mirroring Azure SQL Database
- Mirroring Cosmos DB and Snowflake
- Open mirroring
- Near-real-time replication to OneLake

Copy Job (3)

- Copy job for bulk and incremental
- Supported sources
- Watermark-based incremental copy

34 MODULE 34 · Fabric 5: Real-Time Intelligence

4 SESSIONS | 27 TOPICS | DP-700 ALIGNED

Streaming analytics in Fabric: Eventstream, Eventhouse, KQL and real-time alerts. LAB: Stream card transactions through Eventstream into an Eventhouse, detect anomalies with KQL and fire an Activator alert.

Real-Time Hub & Eventstream (5)

- Real-Time hub
- Eventstream sources: Event Hubs, IoT, CDC
- Eventstream transformations
- Destinations: Eventhouse, Lakehouse, Activator
- Derived streams

Eventhouse & KQL Database (7)

- Eventhouse architecture
- KQL databases and tables
- Ingestion methods
- Native tables vs OneLake shortcuts
- Query acceleration for shortcuts
- Update policies
- Materialized views

KQL (Kusto Query Language) (6)

- where, project, extend
- summarize and bin()
- join and union
- Time-series functions
- Windowing functions
- KQL querysets

7

PHASE 7 OF 7 · MODULES 36–42 · 17 SESSIONS

Streaming, Governance & DataOps

Event Hubs, Purview, Monitor, CI/CD, IaC, FinOps, GenAI

35 MODULE 35 · Fabric 6: Lifecycle, Security & Monitoring

3 SESSIONS | 25 TOPICS | DP-700 ALIGNED

Running Fabric like an enterprise platform: Git, deployments, security layers and monitoring. LAB: Connect a workspace to Git, promote through a deployment pipeline and secure it with OneLake security and RLS.

Lifecycle Management (5)

- Git integration: Azure DevOps and GitHub
- Branching for workspaces
- Deployment pipelines: Dev, Test, Prod
- Deployment rules
- Fabric REST APIs and CLI

Security & Governance (9)

- Workspace-level access controls
- Item-level access controls
- Row, column, object-level security
- Folder and file-level access
- OneLake security
- Dynamic data masking
- Sensitivity labels
- Endorsement: promoted, certified
- Fabric audit logs

Monitoring (6)

- Monitoring hub
- Monitor ingestion and transformation
- Semantic model refresh monitoring
- Alerts
- Capacity Metrics app
- Workspace monitoring

Identify & Resolve Errors (5)

- Pipeline errors
- Dataflow Gen2 errors
- Notebook errors
- T-SQL errors
- OneLake shortcut errors

36 MODULE 36 · Event Hubs & Azure Stream Analytics

3 SESSIONS | 21 TOPICS | DP-700 ALIGNED

Azure-native event ingestion and SQL-based stream processing. LAB: Send 10,000 mock transactions to Event Hubs, aggregate them in 5-minute ASA windows and land them in ADLS and Power BI.

Stream Processing Concepts (5)

- Events and event time
- Processing time
- Watermarks and late arrivals
- At-least-once vs exactly-once
- Lambda vs Kappa on Azure

Azure Event Hubs (6)

- Namespaces and event hubs
- Partitions and throughput units
- Consumer groups
- Capture to ADLS
- Kafka endpoint
- Sending events with the Python SDK

Azure Stream Analytics 6

- Jobs: inputs, query, outputs
- Tumbling windows
- Hopping windows
- Sliding and session windows
- Reference data joins
- Outputs: ADLS, SQL, Power BI, Fabric

Related Services 4

- IoT Hub
- Event Grid
- Cosmos DB change feed
- Choosing ASA vs Spark vs Fabric RTI

37 MODULE 37 · Microsoft Purview: Governance & Cataloguing

2 SESSIONS | 15 TOPICS | DP-700 ALIGNED

Discovering, classifying and tracing data across Azure, Databricks and Fabric (formerly Azure Purview). LAB: Scan your lake and SQL database, auto-classify PII, build a glossary and trace lineage from source to report.

Why Governance 3

- Data governance pillars
- Catalogue, lineage, quality, access
- Regulatory drivers

Unified Catalog 5

- Business glossary
- Governance domains
- Data products
- Data quality rules
- Access requests

Lineage 3

- ADF and Synapse lineage
- Fabric lineage in Purview
- Databricks Unity Catalog integration

Purview Data Map 4

- Registering sources
- Scanning ADLS, Azure SQL, Synapse
- Classifications and sensitivity labels
- Collections and permissions

38 MODULE 38 · Azure Monitor & Log Analytics

2 SESSIONS | 13 TOPICS | DP-700 ALIGNED

Platform-wide observability: one place to see failures, costs and performance. LAB: Build a single workbook showing pipeline failures, durations and storage growth across the platform.

Azure Monitor 5

- Metrics vs logs
- Diagnostic settings
- Activity log
- Action groups
- Metric and log alerts

Log Analytics 4

- Workspaces
- KQL for operations
- Querying ADF, Databricks, Synapse logs
- Saved queries and functions

Dashboards & Workbooks 4

- Azure Workbooks
- Pipeline health dashboard
- SLA tracking
- Integration with Teams

39 MODULE 39 · CI/CD with Azure DevOps & GitHub

3 SESSIONS | 20 TOPICS | DP-700 ALIGNED

Version control and automated deployments across ADF, Databricks, Synapse and Fabric. LAB: One YAML pipeline that tests and deploys ADF, a Databricks bundle and SQL schema to Dev and Prod with approvals.

Git Essentials (5)

- clone, add, commit, push
- Branches and merges
- Pull requests and code review
- Branching strategies: GitFlow, trunk- based
- Resolving conflicts

Azure DevOps (7)

- Azure Repos
- Azure Boards basics
- YAML pipelines
- Stages, jobs, steps
- Service connections
- Variable groups and Key Vault
- Approvals and environments

GitHub Actions (3)

- Workflows and triggers
- Secrets and OIDC to Azure
- Reusable workflows

Data Platform Deployments (5)

- ADF ARM deployments
- Databricks bundles deployment
- Fabric deployment pipelines and Git
- SQL DACPAC deployments
- Testing in CI: pytest, SQL tests

40 MODULE 40 · Infrastructure as Code: Bicep & Terraform

3 SESSIONS | 19 TOPICS | Industry ALIGNED

Creating whole data platforms from code, repeatably, for every environment. LAB: Deploy ADLS, Key Vault, ADF and a Databricks workspace for Dev and Prod with Terraform from a pipeline.

IaC Concepts (4)

- Why infrastructure as code
- Declarative vs imperative
- ARM templates (and why Bicep replaced hand-written ARM)
- State and drift

Terraform (6)

- Providers: azurerm, databricks
- Resources and data sources
- Variables and outputs
- Remote state in Azure Storage
- plan and apply
- Modules and workspaces

Platform Builds
4
- Storage, Key Vault, ADF in code
- Databricks workspace and Unity Catalog via Terraform
- Fabric Terraform provider overview
- IaC in CI/CD pipelines

Bicep
5
- Resources and parameters
- Variables and outputs
- Modules
- Deploying with az deployment
- What-if previews

41 MODULE 41 · Cost Management & FinOps for Data Platforms

1 SESSIONS | 10 TOPICS | Industry ALIGNED

Designing for cost from day one, a skill managers increasingly hire for. LAB: Estimate a monthly bill for the capstone platform and cut it by 30% with design changes.

Azure Cost Management
5
- Cost analysis
- Budgets and alerts
- Tagging for chargeback
- Reservations and savings plans
- Pricing calculator

Service Cost Levers
5
- Storage tiers and lifecycle
- ADF DIUs and activity runs
- Databricks DBUs and compute choice
- Synapse pause/resume
- Fabric capacity sizing and smoothing

42 MODULE 42 · GenAI for Data Engineers

3 SESSIONS | 17 TOPICS | Industry ALIGNED

Preparing data for AI and using AI to engineer data faster. LAB: Build a pipeline that chunks and embeds policy PDFs into Azure AI Search and answer questions with a RAG endpoint.

GenAI Foundations
4
- LLMs, tokens and embeddings
- Prompting basics for engineers
- Azure OpenAI in Microsoft Foundry
- Responsible AI

Vectors & Retrieval
5
- Embeddings pipelines
- Chunking strategies
- Azure AI Search: vector and hybrid search
- Integrated vectorisation
- RAG architecture

AI-Ready Data Pipelines
4
- Processing PDFs and documents
- Document Intelligence overview
- Metadata enrichment
- Refresh and re-index pipelines

AI as Your Co-Engineer
4
- Copilot in Fabric
- Databricks Assistant
- GitHub Copilot for PySpark and SQL
- Natural-language-to-SQL and its limits

OPTIONAL ADD-ON · 6 SESSIONS

Snowflake on Azure

For learners targeting Snowflake-based teams

A1 MODULE A1 · Optional Add-on: Snowflake on Azure

6 SESSIONS | 22 TOPICS | SnowPro Core ALIGNED

For learners targeting Snowflake shops, run on Azure with ADLS integration. LAB: Auto-ingest ADLS files with Snowpipe, transform with streams and tasks and share a secure view.

Snowflake Architecture (4)

- Storage, compute, cloud services layers
- Virtual warehouses
- Editions and Azure regions
- Snowsight UI

Transformations (5)

- Snowflake SQL
- Semi-structured data: VARIANT, FLATTEN
- Streams and tasks
- Dynamic tables
- Stored procedures and Snowpark Python

Platform Features (5)

- Time Travel and Fail-safe
- Zero-copy cloning
- Secure data sharing
- RBAC and masking policies
- Resource monitors and cost

Loading Data from Azure (5)

- Storage integrations with ADLS
- External stages
- File formats
- COPY INTO
- Snowpipe with Event Grid

Integrations (3)

- ADF and Fabric connectors
- Fabric mirroring for Snowflake
- Iceberg tables overview

CAPSTONE PROJECTS

Four projects. Four industries. One portfolio.

Each capstone is built end to end in your own Azure and Fabric environment, published on GitHub with an architecture diagram and README, and presented in a live demo with trainer review.

CAPSTONE01 · RETAIL & E-COMMERCE

Retail Lakehouse on Azure Databricks

An enterprise lakehouse that unifies orders, customers, products and inventory from multiple systems.

ADF ADLS Gen2 Auto Loader Lakeflow Pipelines Unity Catalog Delta Lake Power BI

1. Metadata-driven ADF ingestion from Azure SQL and
2. Auto Loader into Bronze with schema evolution
3. Silver cleansing with expectations and SCD2 product
4. Gold sales, inventory and customer KPIs
5. Unity Catalog masking, row filters and lineage
6. Power BI dashboard on a SQL warehouse

CAPSTONE02 · BANKING & PAYMENTS

Real-Time Fraud Monitoring

Card transactions are scored in seconds and suspicious patterns trigger live alerts.

Event Hubs Fabric Eventstream Eventhouse / KQL Stream Analytics Activator Real-Time Dashboard

1. Python producer streaming card transactions to Event
2. Eventstream routing to Eventhouse and Lakehouse
3. KQL rules for velocity and geo anomalies
4. ASA tumbling-window aggregates
5. Activator alerts to Teams and email
6. Real-time dashboard for the fraud team

CAPSTONE03 · HEALTHCARE & INSURANCE

Enterprise Warehouse on Microsoft Fabric

A governed, Dev-to-Prod Fabric warehouse for claims analytics with full security.

Fabric Mirroring Lakehouse Warehouse Dataflow Gen2 Deployment Pipelines OneLake Security Purview

1. Mirror the claims database into OneLake
2. Lakehouse Silver layer in notebooks
3. Star schema in Fabric Warehouse with SCD2 stored
4. RLS, column masking and sensitivity labels
5. Git integration and Dev → Test → Prod deployment
6. Purview catalogue and lineage

CAPSTONE04 · CROSS-INDUSTRY DATAOPS

Platform-as-Code with an AI Data Assistant

The whole platform built from code, deployed by CI/CD, monitored, cost-controlled and AI-enabled.

Terraform / Bicep Azure DevOps / GitHub Actions ADF Databricks Bundles Azure Monitor Azure OpenAI Azure AI Search

1. Terraform builds storage, Key Vault, ADF and
2. YAML pipeline tests and deploys all components
3. Log Analytics workbook and alerting
4. Budgets and tagging for cost control
5. RAG assistant over the data catalogue and docs
6. Architecture review and live demo

Build a practical portfolio

The program's four end-to-end capstone builds across retail, banking, healthcare and DataOps.

01

Retail lakehouse on Azure

Ingest retail data with ADF into ADLS Gen2, process with Auto Loader and Lakeflow pipelines on Databricks.

02

Real-time fraud monitoring

Stream banking events with Event Hubs and Stream Analytics, with live dashboards and Activator alerts.

03

Fabric data warehouse

Model a healthcare claims warehouse in Microsoft Fabric with T-SQL and serve governed marts.

04

Platform-as-Code with AI

Deploy the whole platform with Bicep, Terraform and CI/CD, then add an Azure OpenAI data assistant.

Your portfolio tells the story

Show the pipeline design, code, data quality checks and dashboard. Explain the business question and your technical choices.

One stack. Four certification paths.

Use the course topics as a starting point for further certification study. Each provider sets its own exam requirements.

01

Microsoft Fabric Data Engineer (DP-700)

OneLake, lakehouses, warehouses, real-time intelligence and pipelines in Microsoft Fabric.

02

Microsoft Azure Data Fundamentals (DP-900)

Core data concepts and Azure data services: ADLS Gen2, databases and analytics.

03

Microsoft Power BI Data Analyst (PL-300)

Data preparation, data modelling, DAX and reporting in Power BI.

04

Databricks Certified Data Engineer Associate

Further study in Spark and lakehouse concepts, building on the Azure Databricks modules.

Course completion certificate

Complete the course to earn a Greens Technologies certificate of completion.

Provider certification exams are optional. Exam fees are separate. The course does not award these provider credentials automatically.

A practical data engineering toolkit

Learn how each tool fits into one connected workflow, from source data to business reporting.

SQL

Query, join and model business data.

Python

Read files and APIs; build reusable data scripts.

Azure

ADLS Gen2, Entra ID, Key Vault and Purview.

Data Factory

Build, run and monitor data pipelines.

Databricks

Spark, Delta Lake and Unity Catalog.

Microsoft Fabric

OneLake, lakehouse, warehouse, real-time intelligence.

Power BI

Create data models, reports and dashboards.

ALSO IN THE LABS

Azure Synapse, Event Hubs, Git, Azure DevOps, Terraform, Bicep, Purview and Azure OpenAI.

The end-to-end flow

Source data > ADLS Gen2 > Data Factory > Databricks > Fabric > Power BI

Where data skills can take you

Build toward data engineering and analytics roles with a connected cloud data portfolio.

Career path

Junior Data Engineer > Data Engineer > Senior Data Engineer > Lead Data Engineer

Salary range

Rs 10L to 45+L

per year

Indicative salary range. Actual offers vary by role and experience.

Companies hiring data engineering talent

PRODUCT-BASED

SERVICE-BASED

Company names show examples of employers for these skills, not a list of placement partners or a promise of placement at a specific company.

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PLACEMENTS GUARANTEED

Build your skills and prepare for the next step with Greens Technologys.

01

Resume preparation

Present your tools, projects and skills clearly, with trainer review.

02

Mock interviews

Practise technical answers and project explanations, then use the feedback.

03

Job-ready profiles

Create and improve your Naukri and LinkedIn profiles.

04

Interview coordination

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